Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Autoregressive Moving Average Model

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https://sithuaung-ink.github.io/data-analyst-portfolio/

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What is ARMA?

- ARMA stands for Autoregressive Moving Average (ARMA) model.
- ARMA models are formed by combining the AR and MA models.
- An AR or MA model alone to fit the time series is sometimes difficult to capture the complex structure of this time series.
- ARMA is used to model stationary time series (no trend, no seasonality).

AR(p) – Autoregressive Model

Autoregressive(AR) can be denoted as

$$y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + \epsilon_t$$

where p is number of lagged values, φ is the coefficient and ϵ_t is the white error.

If $p = 1$, $\varphi = 0.7$

$$y_t = 0.7 y_{t-1} + \epsilon_t$$

MA(q) – Moving Average Model

Moving Average (MA) can be defined as follows

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ \dots + \theta_q \epsilon_{t-p}$$

where q is the number of lagged errors used, θ is the coefficient and ϵ_t is the white error.

If q = 1, θ= 0.5

$$y_t = \epsilon_t + 0.5\epsilon_{t-1}$$

Autoregressive Moving Average Model

The definition of ARMA models as follows.

$$y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-p}$$

ARMA combines both past values and past errors.

How to Choose p and q

For AR(p):

- PACF cuts off at lag p
- ACF tails off – slowly decays
- For e.g., PACF is significant at lag 1 only, choose $p = 1$

For MA(q):

- ACF cuts off at lag q
- PACF tails off – slowly decays
- For e.g., ACF is significant at lag 2 only, choose $q = 2$

How to Choose p and q

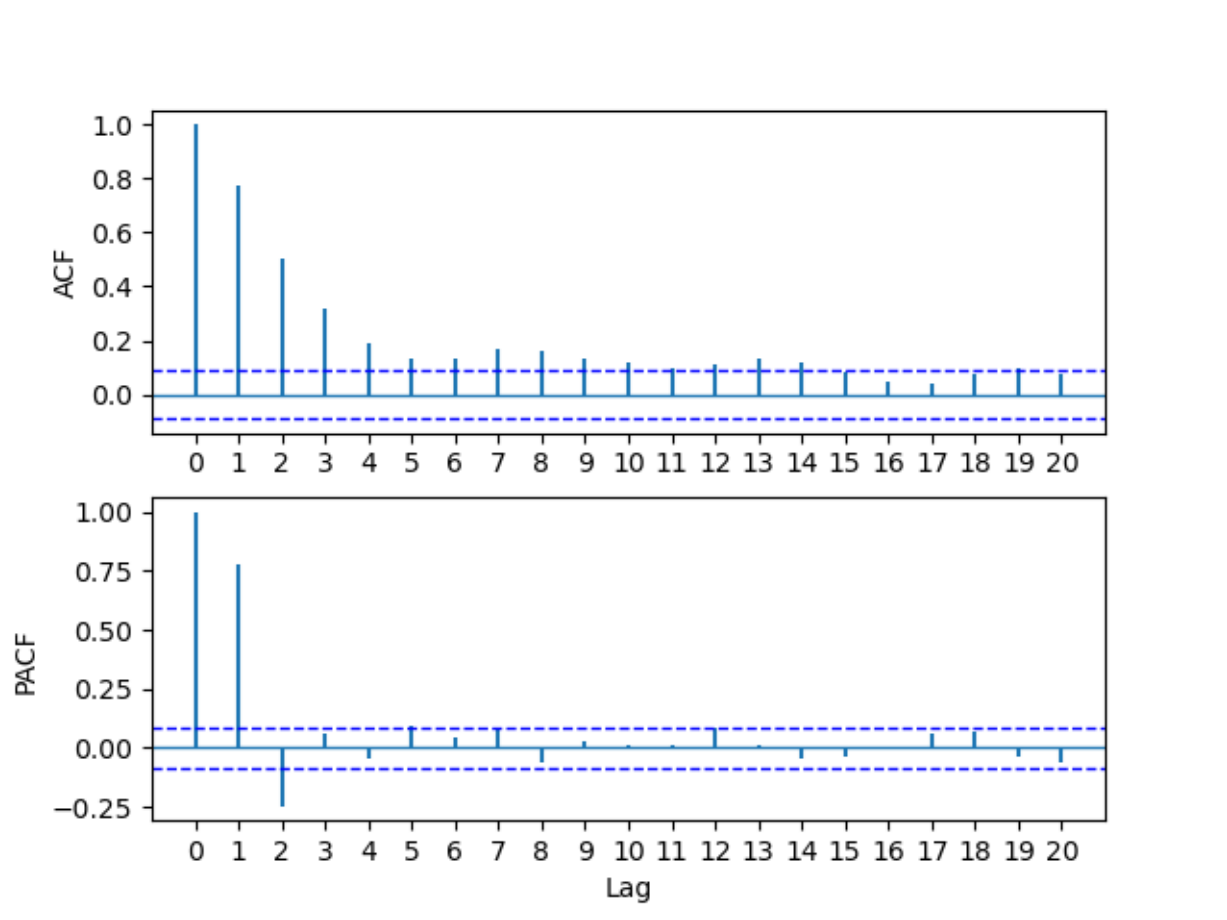
ACF and PACF tails off, and No sharp cutoff

Try a manual combination of

- ARMA (1,0),
- ARMA (0,1),
- ARMA (1,1)

Then more parameters to consider

- AIC
- BIC (or HQIC).



Order Determination

Akaike Information Criterion (AIC)

AIC = –2 (maximized log likelihood) + 2 (No. of estimated parameters)

Bayesian Information Criterion (BIC)

BIC = –2 (maximized log likelihood) + log(n) (No. of estimated parameters)

Hannan-Quinn Information Criterion (HQIC)

HQIC = –2 (maximized log likelihood) + log log(n) (No. of estimated parameters)

All the values of AIC, BIC, and HQIC are smaller-the-better. We will use the Python function to calculate these values.

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Stationarity Test

KPSS Test (Kwiatkowski-Philips-Schmidt-Shin)

Null hypothesis:

H_0 : Time Series is stationary

Alternative

H_1: Time series is non-stationary

Interpretation:

p-value < 0.05 -> reject H_0 -> Time Series is non-stationary

p-value > 0.05 -> fail to reject -> Time series is Stationary

Forecasting

Two types of forecast,

- One is in-sample prediction: the trained model uses data from 2010-2020, and the predicted values are also within 2010-2020.
- The other is out-of-sample: predict on new data not used in training. Train the model on 2010-2020 and predict 2021-2022.
- We will use in-sample training (i.e., train data on 2010-2020) and out-of-sample to predict (i.e., 2020 and future years for prediction).

ARIMA (1,0,1)

Simulating and building an ARMA (1,1) model

$$y_t = \varphi_1 y_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

If $\varphi = 0.6$, $\theta = 0.4$

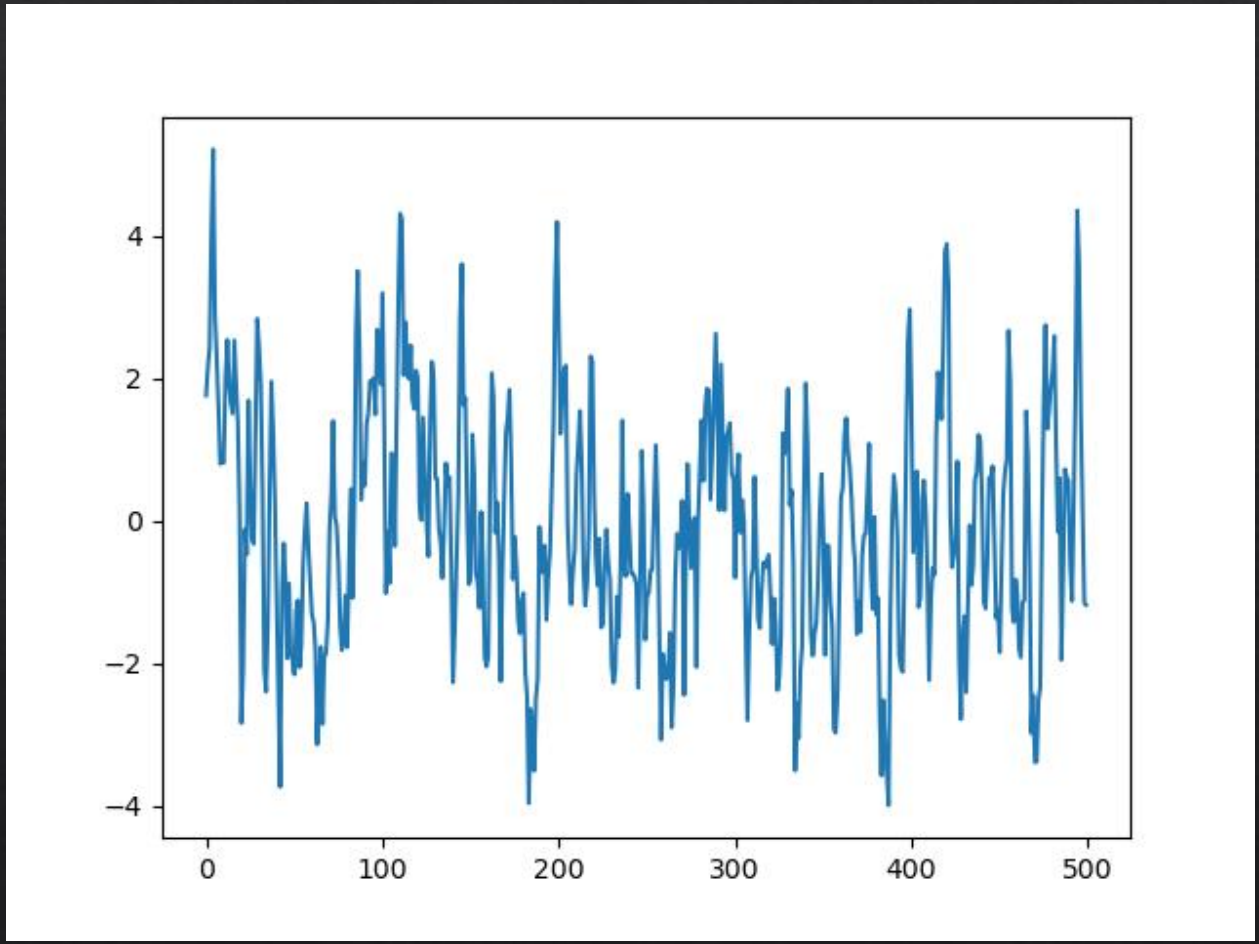
$$y_t = 0.6 y_{t-1} + \epsilon_t + 0.4 \epsilon_{t-1}$$

Generate this series in Python.

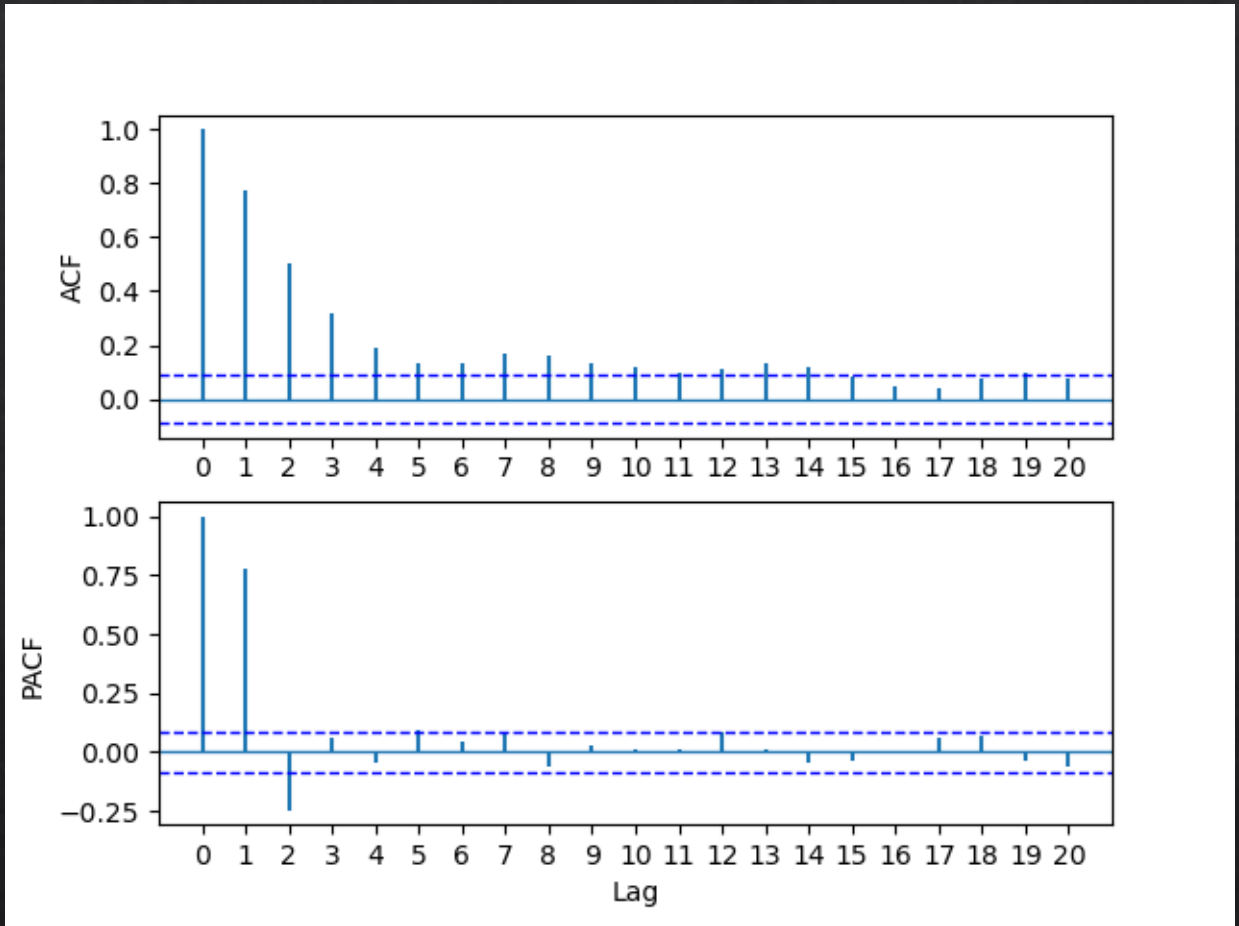
bash

```bash
pip install PythonTsa
```

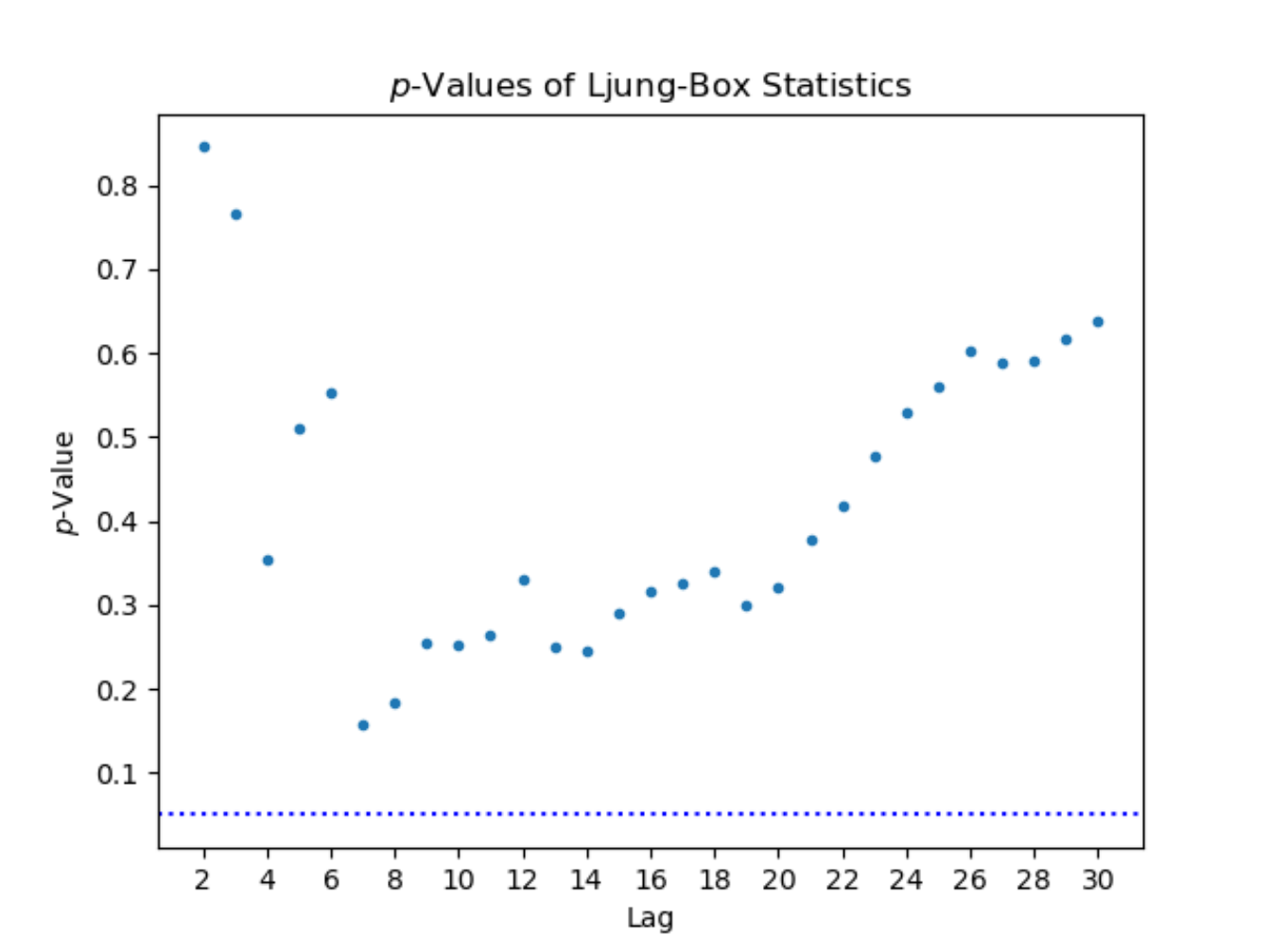
ARIMA (1,0,1)



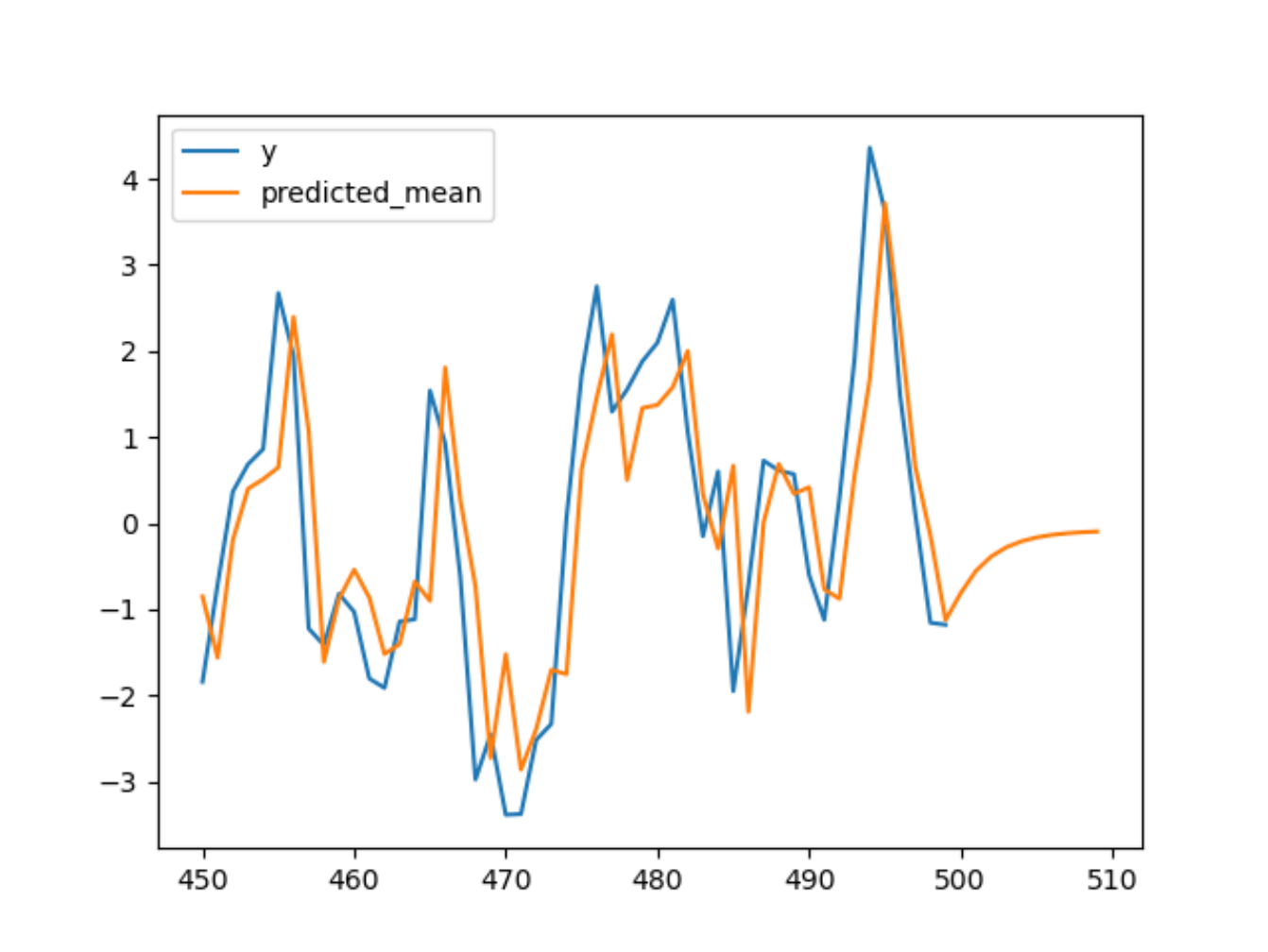
ARIMA (1,0,1)


ACF
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
PACF
1.00
0.75
0.50
0.25
0.00
−0.25
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Lag

ARIMA (1,0,1)


p-Values of Ljung-Box Statistics
p-Value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
Lag

ARIMA (1,0,1)



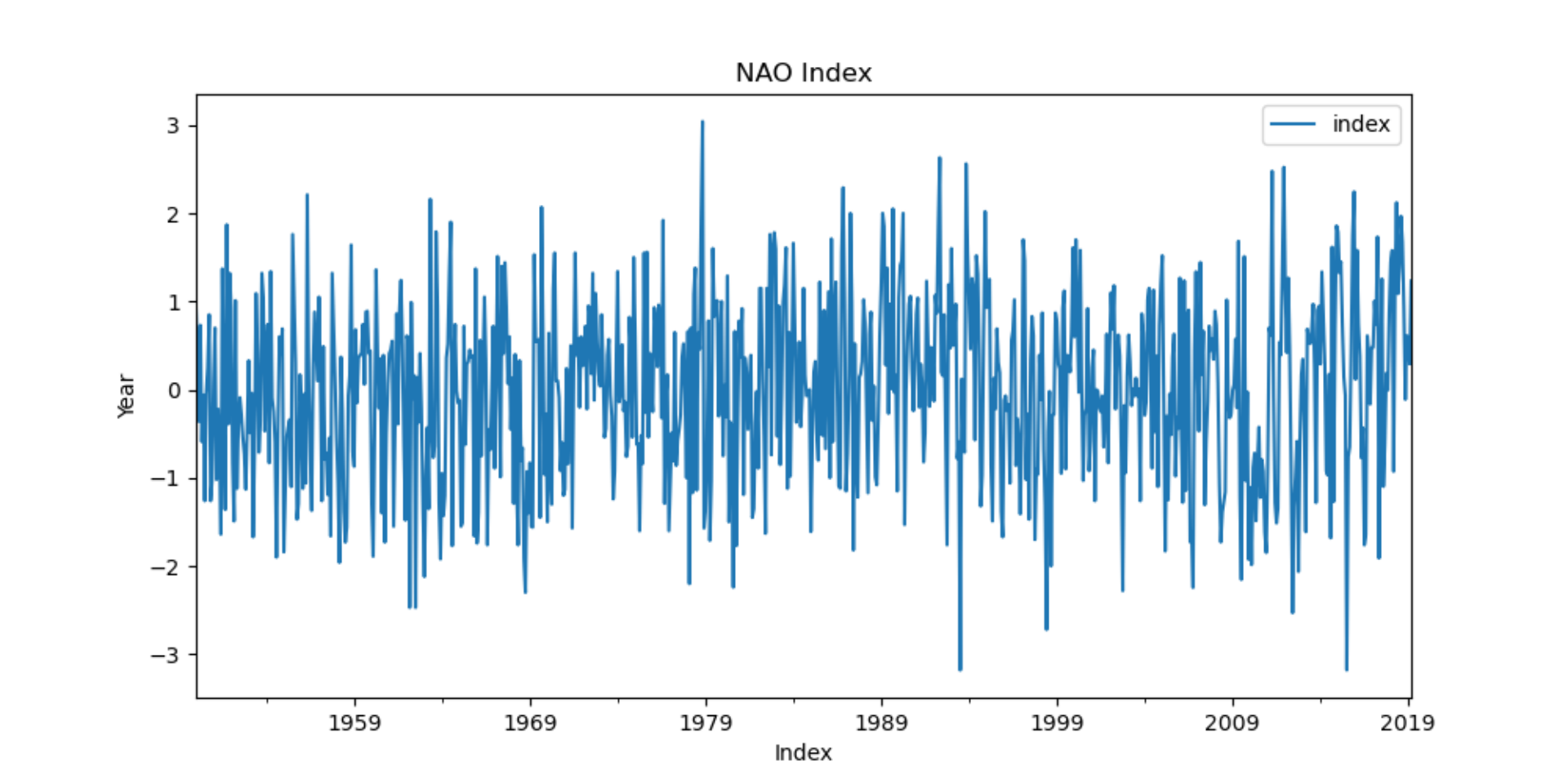
NAO index

Dataset description

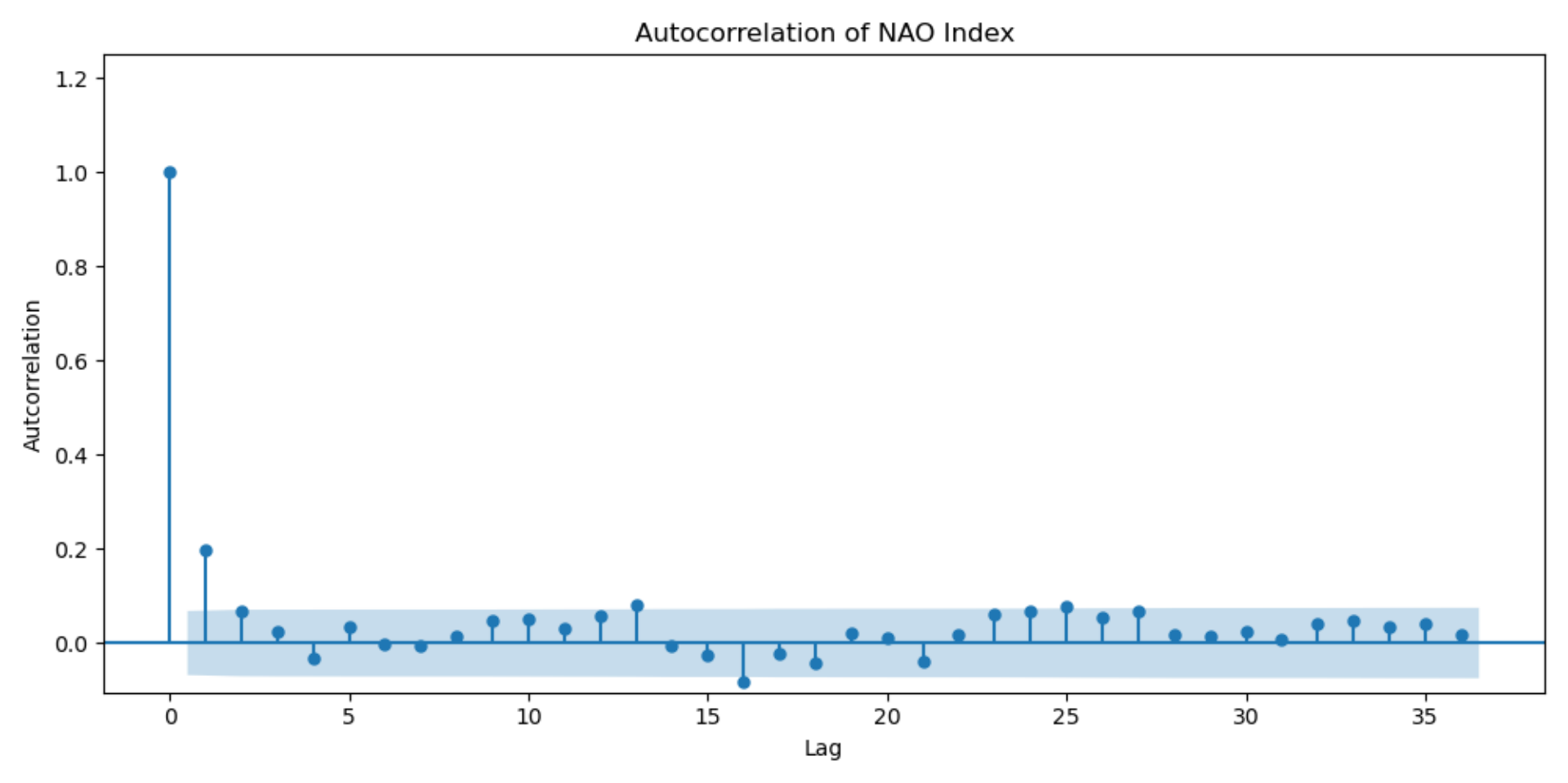
- The North Atlantic Oscillation (NAO) is the monthly mean NAO index since January 1950.
- The NAO index is important for weather research and is based on the surface sea level pressure difference between the subtropical high and the subpolar low.
- A positive index brings mild, wet winters to Europe and warm, dry winters to the U.S. East, while a negative index causes cold, snowy conditions in both regions.[1]

[1]https://www.climate.gov/news-features/understanding-climate/climate-variability-north-atlantic-oscillation

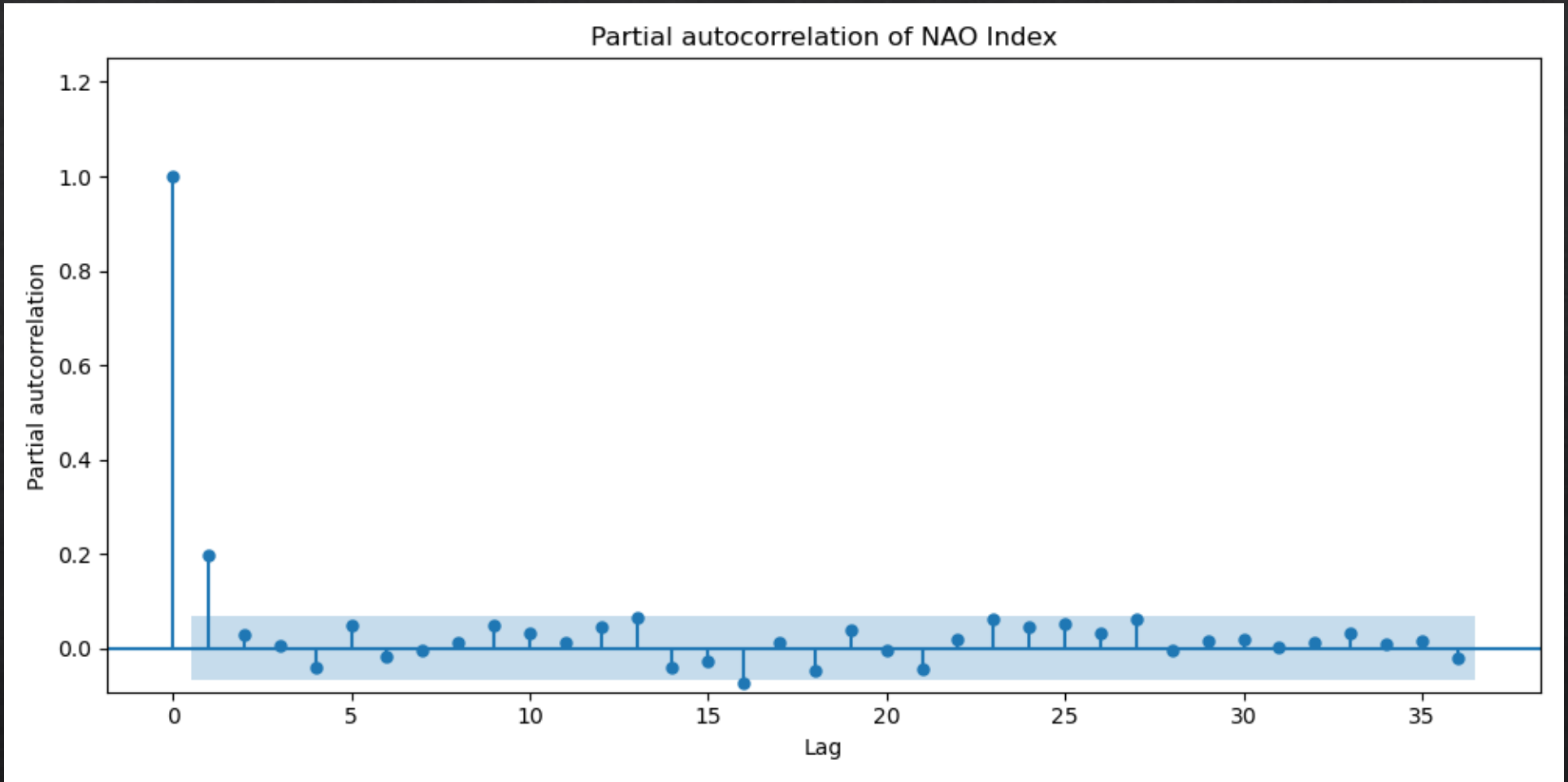
NAO index


NAO Index
index
3
2
1
0
−1
−2
−3
Year
1959
1969
1979
1989
1999
2009
2019
Index

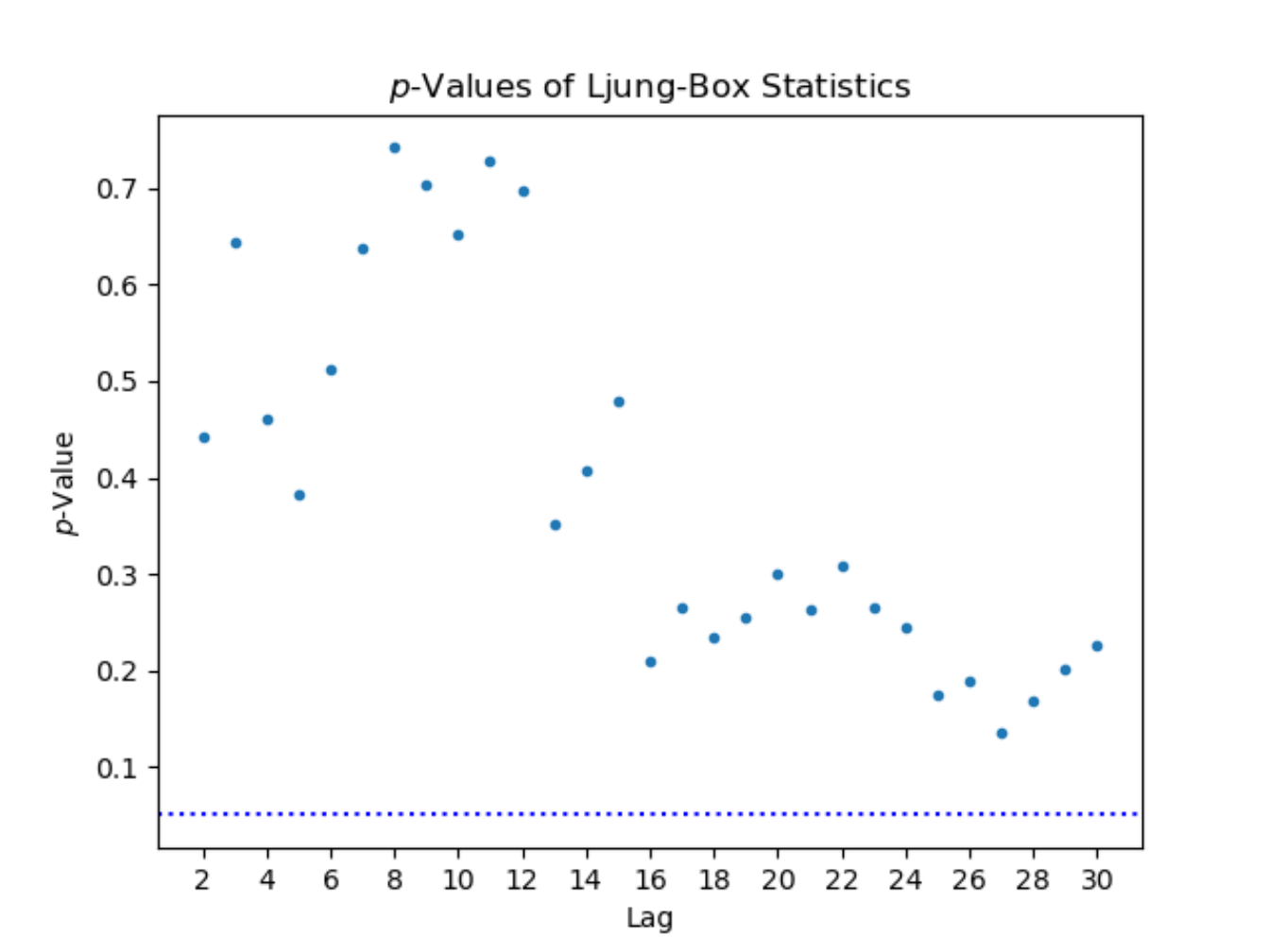
NAO index


Autocorrelation of NAO Index
Autcorrelation
Lag
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
35

NAO index


Partial autocorrelation of NAO Index
Partial autcorrelation
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
35
Lag

NAO index


p-Values of Ljung-Box Statistics
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0.7
0.6
0.5
0.4
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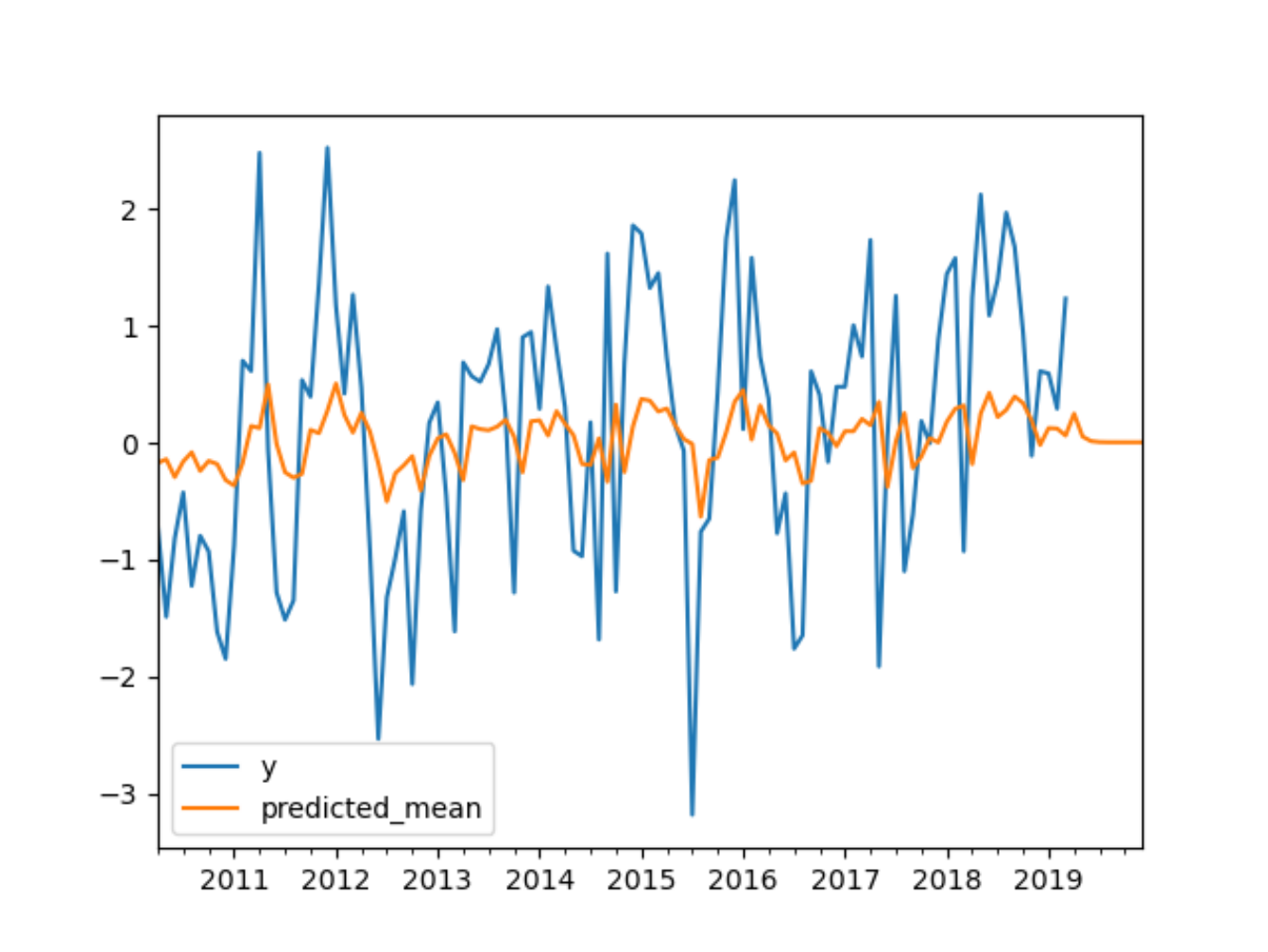
0.2
0.1
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
Lag

NAO index


2
1
0
−1
−2
−3
2011
2012
2013
2014
2015
2016
2017
2018
2019
y
predicted_mean